Final Project Checkpoint 2

CMSC 320 - Introduction to Data Science
Fall 2026
Instructor: Dr. Fardina Alam

Summary

For your final semester long group project, you will publish a tutorial that will walk users through the entire data science pipeline: acquiring and curating data, parsing the data into a queryable format, exploratory data analysis, hypothesis testing and machine learning, and explaining the results with words and visualizations. This project is designed to demonstrate your data science knowledge and contribute to your portfolio.

Deadlines

- **Project posted:** September 8, 2026
- **First deliverable:** September 29, 2026
- **Second deliverable:** October 29, 2026
- **Final deliverable:** December 10, 2026

Instructions for later deliverables will be posted no later than 3 weeks before the date they are due. Late submissions will **not** be accepted for the final deliverable.

Rubrics

The final project is graded out of 100 points. Each deliverable must be submitted on **Gradescope as a group**. Working alone is permitted. We recommend four people per group. The maximum group size is 6 people.

IMPORTANT: If any kind of plagiarism is found, including the use of AI tools like ChatGPT or direct code from previous semesters, you will receive a grade of 0 for the final tutorial. To cite the python libraries you use, simply include the import code block at the top of the tutorial.

Checkpoint 2 (25 points)

Submit a **jupyter notebook file (.ipynb)** on Gradescope with:

- **(5 points) Data preprocessing:** (a) import, (b) parse (e.g., convert strings to ints), (c) organize (e.g., set up a database or a pandas DataFrame).
- **(20 points) Basic data exploration and summary statistics**
    - You must present three conclusions using at least three different statistical methods including hypothesis testing.

        - For example: What are the main characteristics of your dataset? How many features and entries are there? Is a feature over-represented? Are features correlated? Are there outliers? Identify the attributes that will affect your choice of primary analysis technique. Etcetera.
    - For each method, you must have at least one **gorgeous** plot.

Checkpoint 3 - Final deliverable (70 points)

Submit a **single URL** pointing to your final tutorial to Gradescope. The tutorial should be self-contained, a mix of Markdown prose and Python code, and delivered as a GitHub statically-hosted Page (see the Publishing section below for instructions). The format of the final tutorial is given at the end of this section. **We will release further instructions later in the semester.**

Publishing

GitHub provides a service called Pages (https://pages.github.com/) that provides website hosting functionality backed by a GitHub-based git repository. We would like you to host your final project on a GitHub Pages project site. To do this, you will need to:

1. Create a GitHub account (or use the one you already have) with username <username>.
2. Create a git repository titled username.github.io; make sure username is the same as whatever you chose for your global GitHub account.
3. Create a project within this repository. This is where you'll dump your iPython notebook file and an HTML export of that notebook file.

These instructions are also given on the front page of https://pages.github.com/ and Annie Zhou, a past student, created a tutorial: https://azhou4847.github.io/CMSC320ProjectPublication/. The deliverable to the CMSC320 staff will then be a single URL pointing to this publicly-hosted GitHub Pages-backed website.

Previous Examples

Here are some examples of good final tutorials from previous sections of CMSC 320:

- Machine Learning the Unspoken: Acoustic Feature Analysis for Classifying Intent in Nonverbal Vocalizations, https://visheshnarayan.github.io/
- Predicting House Prices in Ames, Iowa, https://joshua-vallabhaneni.github.io/
- The Effect of Storms in the United States, https://shahsean.github.io/
- An Evaluation of American Presidential Elections, https://jcurran0499.github.io/
- Analysis of S&P 500 Companies, https://neo-zhao.github.io/
- Does the University of Maryland Computer Science Department need more faculty?, https://krixly.github.io/
- Analysis of Crime Data at UMD, College Park, https://andresgogo.github.io/

DO NOT COPY THEIR PROJECTS OR CODE.